

Accelerated Gradient Descent através do espelho ou como acelerar o Mirror Descent

José Arthur de Souza

A Differential Equation for Modeling Nesterov’s Accelerated Gradient Method: Theory and Insights

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Abstract

We derive a second-order ordinary differential equation (ODE) which is the limit of Nesterov’s accelerated gradient method. This ODE exhibits approximate equivalence to Nesterov’s scheme and thus can serve as a tool for analysis. We show that the continuous time ODE allows for a better understanding of Nesterov’s scheme. As a byproduct, we obtain a family of schemes with similar convergence rates. The ODE interpretation also suggests restarting Nesterov’s scheme leading to an algorithm, which can be rigorously proven to converge at a linear rate whenever the objective is strongly convex.

Keywords: Nesterov’s accelerated scheme, convex optimization, first-order methods, differential equation, restarting

1. Introduction

In many fields of machine learning, minimizing a convex function is at the core of efficient model estimation. In the simplest and most standard form, we are interested in solving

$$\text{minimize } f(x),$$

where f is a convex function, smooth or non-smooth, and $x \in \mathbb{R}^n$ is the variable. Since Newton, numerous algorithms and methods have been proposed to solve the minimization problem, notably gradient and subgradient descent, Newton’s methods, trust region methods, conjugate gradient methods, and interior point methods (see e.g. Polyak, 1987; Boyd and Vandenberghe, 2004; Nocedal and Wright, 2006; Ruszczyński, 2006; Boyd et al., 2011; Shor, 2012; Beck, 2014, for expositions).

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Capítulo 1

Introdução

Neste documento consideramos problemas de otimização da forma

$$(P) = \begin{cases} \min f(x) \\ x \in \mathcal{X} \subseteq \mathbb{R}^n \end{cases}, \quad (P)$$

sendo $f : \mathbb{R}^n \rightarrow \mathbb{R}$ uma função diferenciável, convexa com gradiente ∇f L -Lipschitz e $\mathcal{X}$ um conjunto convexo.

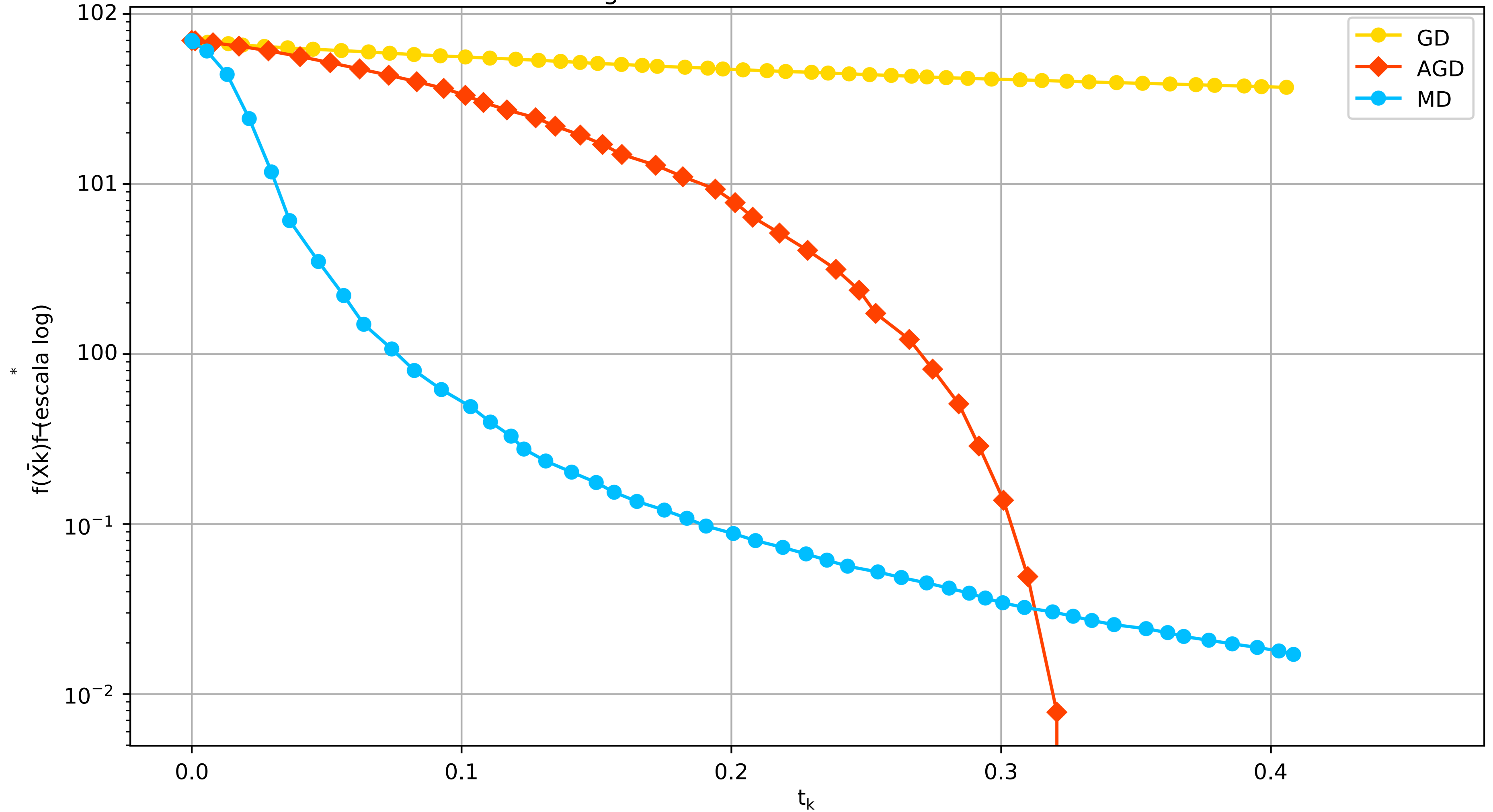
Em otimização convexa, especialmente em contextos de larga escala como em Aprendizado de Máquina, métodos de primeira ordem são fundamentais para a resolução aproximada de tais problemas. Essa relevância deve-se ao baixo custo associado a eles (sendo o cálculo de uma única derivada tipicamente a operação mais custosa computacionalmente) e à possibilidade de paralelização.

Neste trabalho estudamos três métodos de primeira ordem. O primeiro é possivelmente o método de primeira ordem mais simples conhecido: o método do gradiente descendente ou *Gradient Descent* (GD). O GD executa a cada iteração um passo na direção oposta a do gradiente, que representa a direção de maior

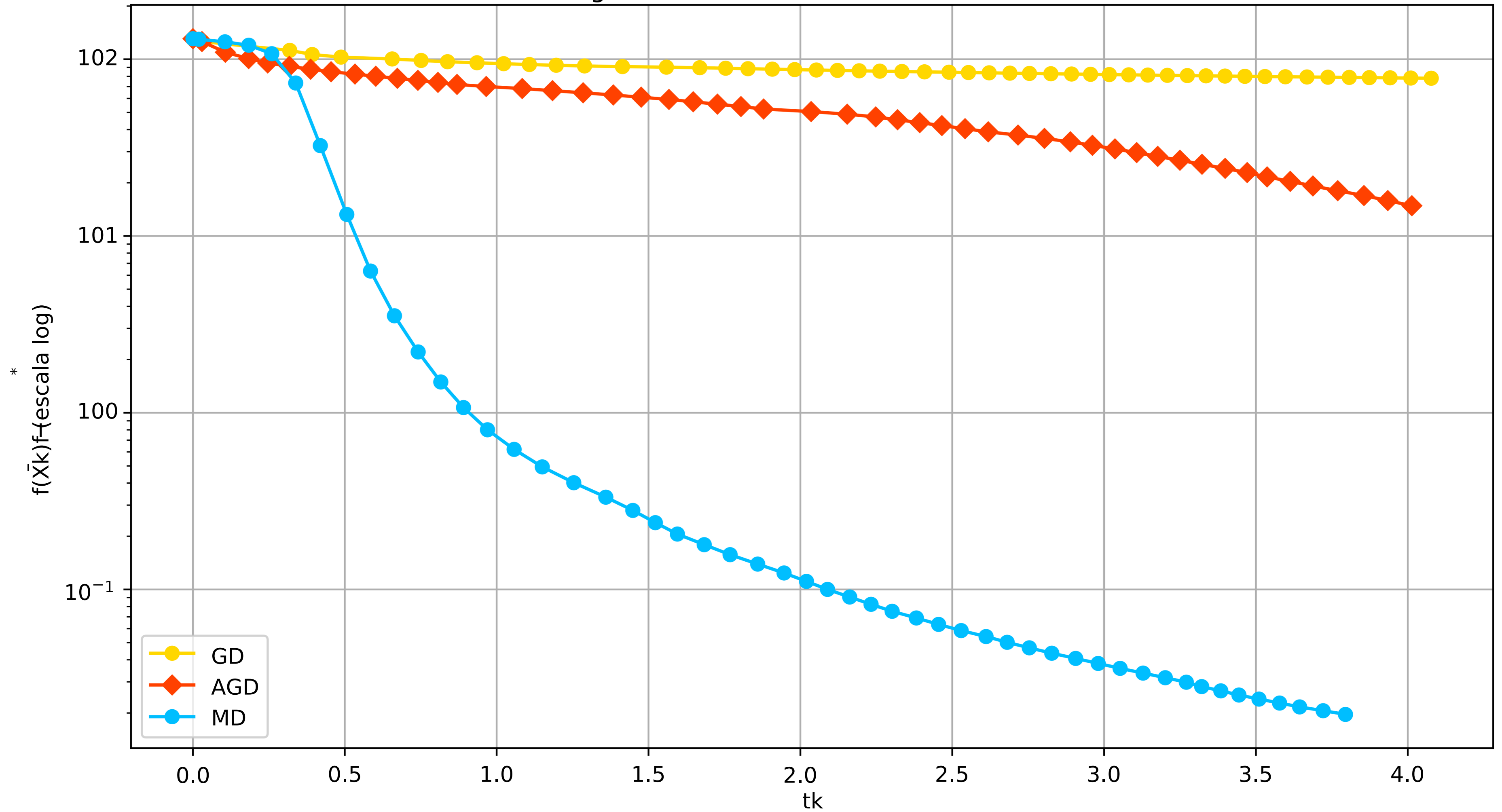
Metodos de primeira ordem:

- Gradient Descent (GD) ou Gradiente Descendente;
- Accelerated Gradient Descent (AGD) ou Método de Nesterov;
- Mirror Descent (MD) ou Método de Bregman.

Convergência dos métodos em dimensão 100



Convergência dos métodos em dimensão 10000



Gradient Descent

$$x_{k+1} = \arg \min_x \left\{ \eta \langle \nabla f(x_k), x \rangle + \frac{1}{2} \|x - x_k\|_2^2 \right\}$$

$$x_{k+1} = x_k - \eta \nabla f(x_k)$$

Gradient Descent

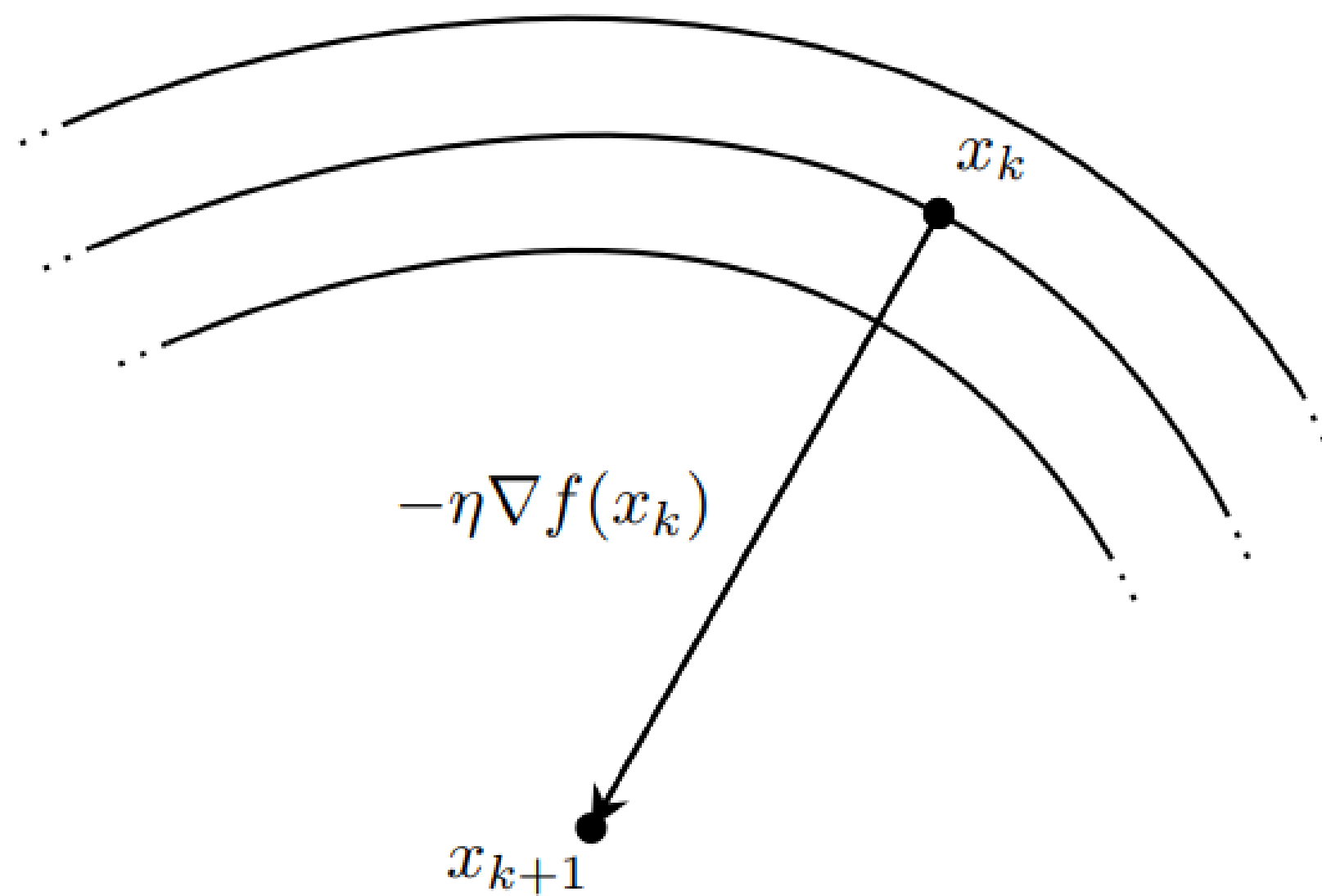
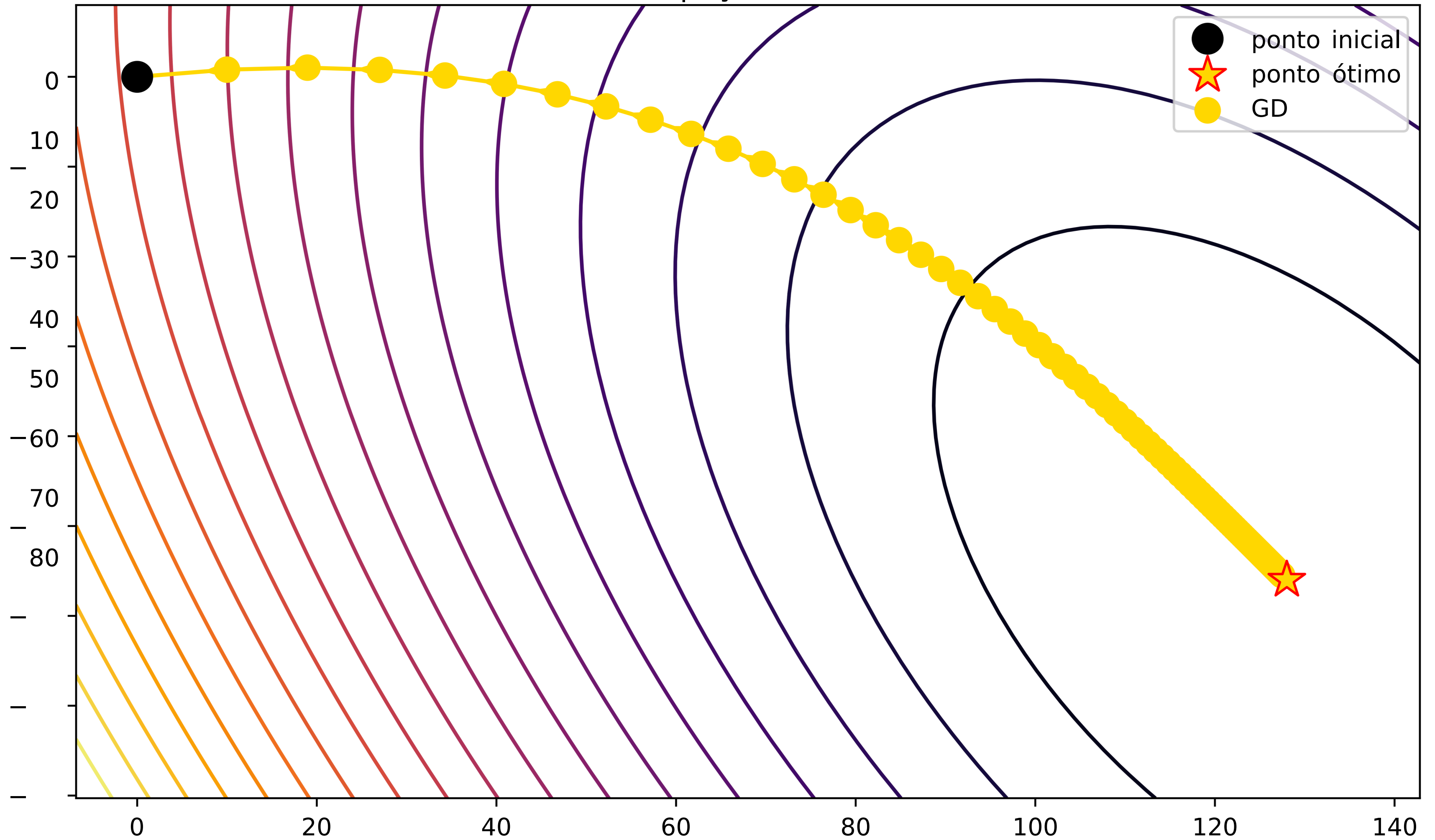


Figura 3.1: Uma iteração do método do gradiente

Espaço Primal



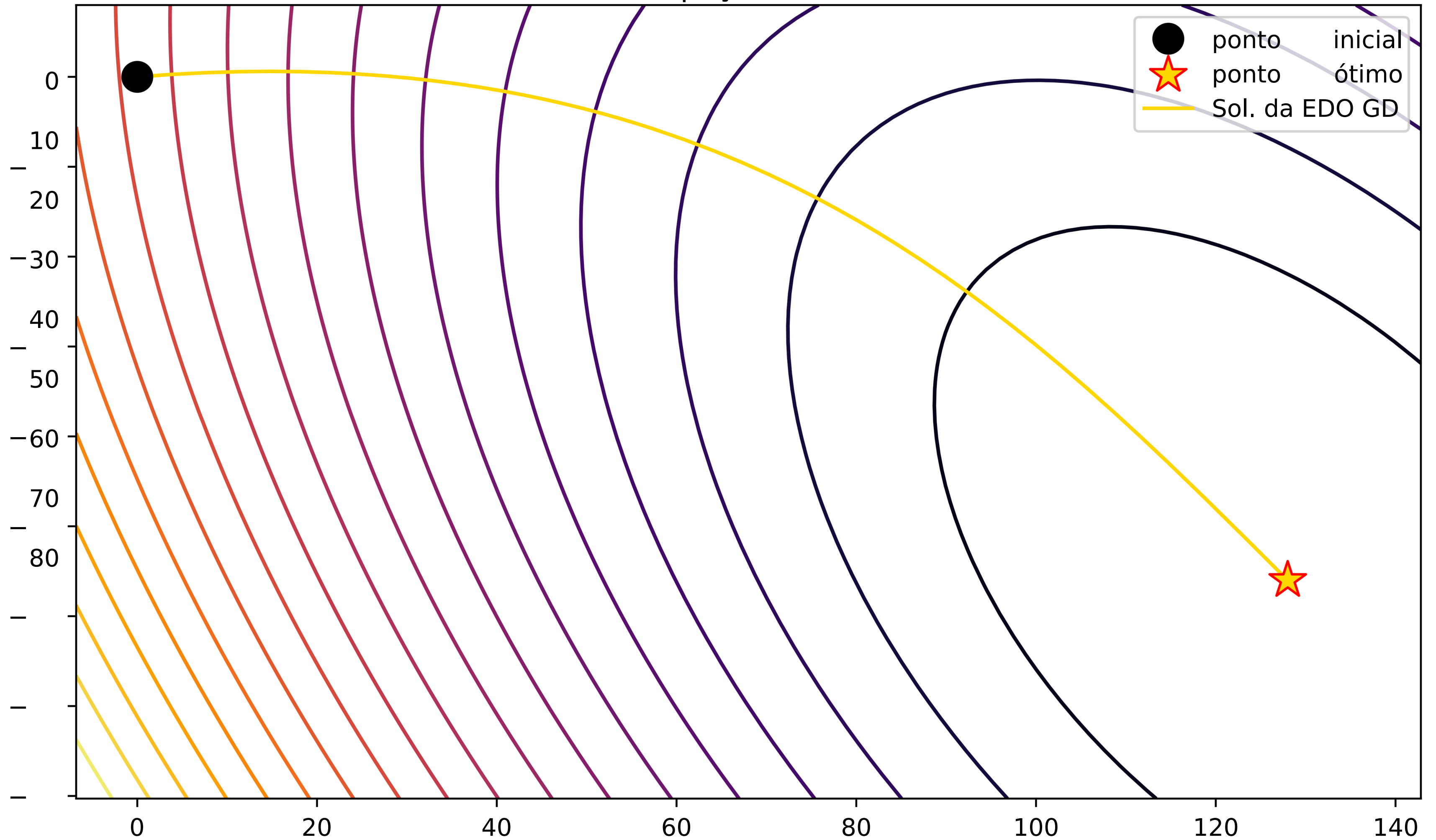
Gradient Descent

$$x_{k+1} = \arg \min_x \left\{ \eta \langle \nabla f(x_k), x \rangle + \frac{1}{2} \|x - x_k\|_2^2 \right\}$$

$$x_{k+1} = x_k - \eta \nabla f(x_k)$$

$$\frac{d}{dt} X(t) = -\nabla f(X(t))$$

Espaço Primal



Gradient Descent

Accelerated Gradient Descent

$$\begin{cases} x_{k+1} &= y_k - \eta \nabla f(y_k) \\ y_k &= x_k + \mu_k (x_k - x_{k-1}) \end{cases}$$

Accelerated Gradient Descent

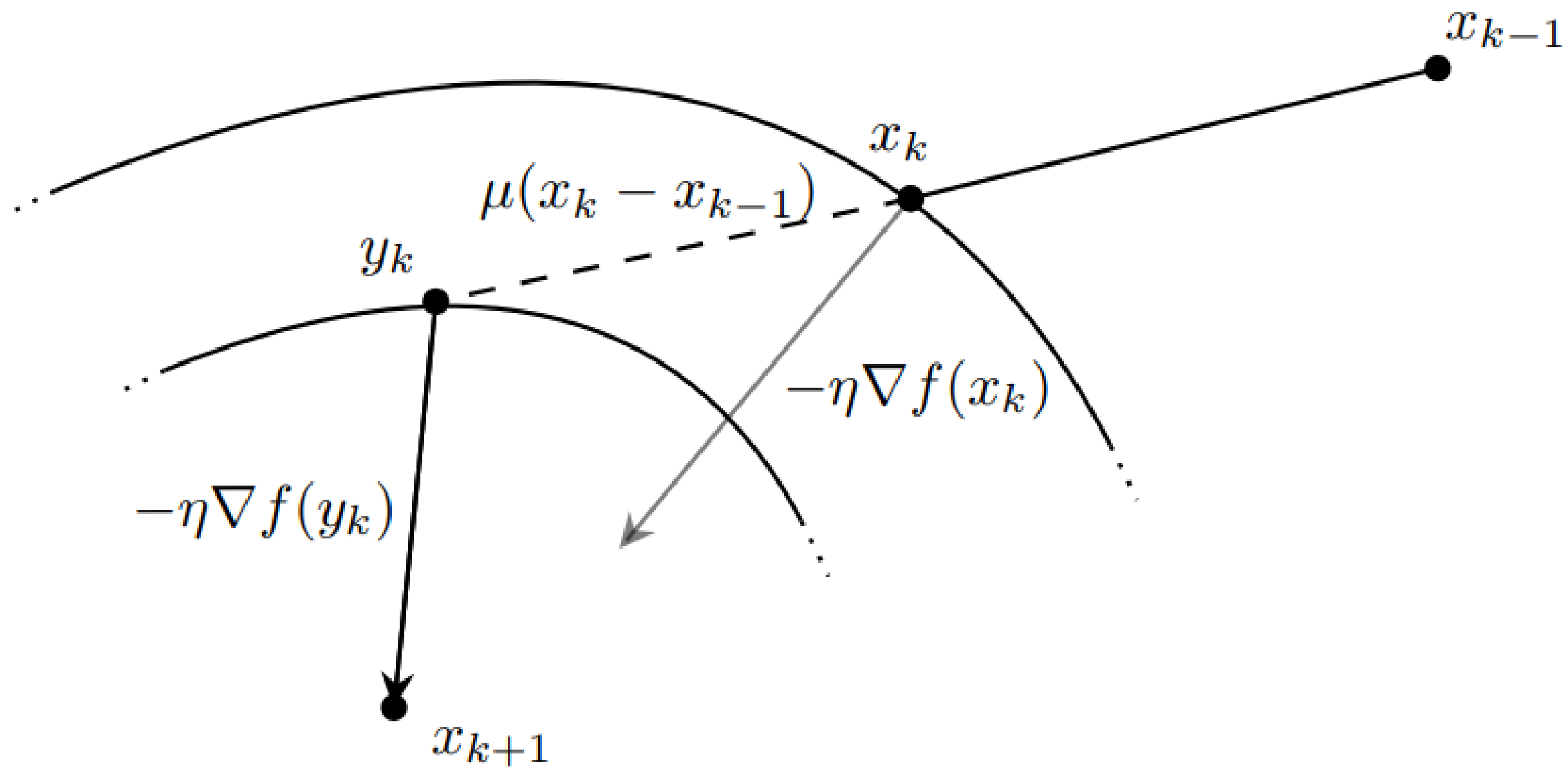
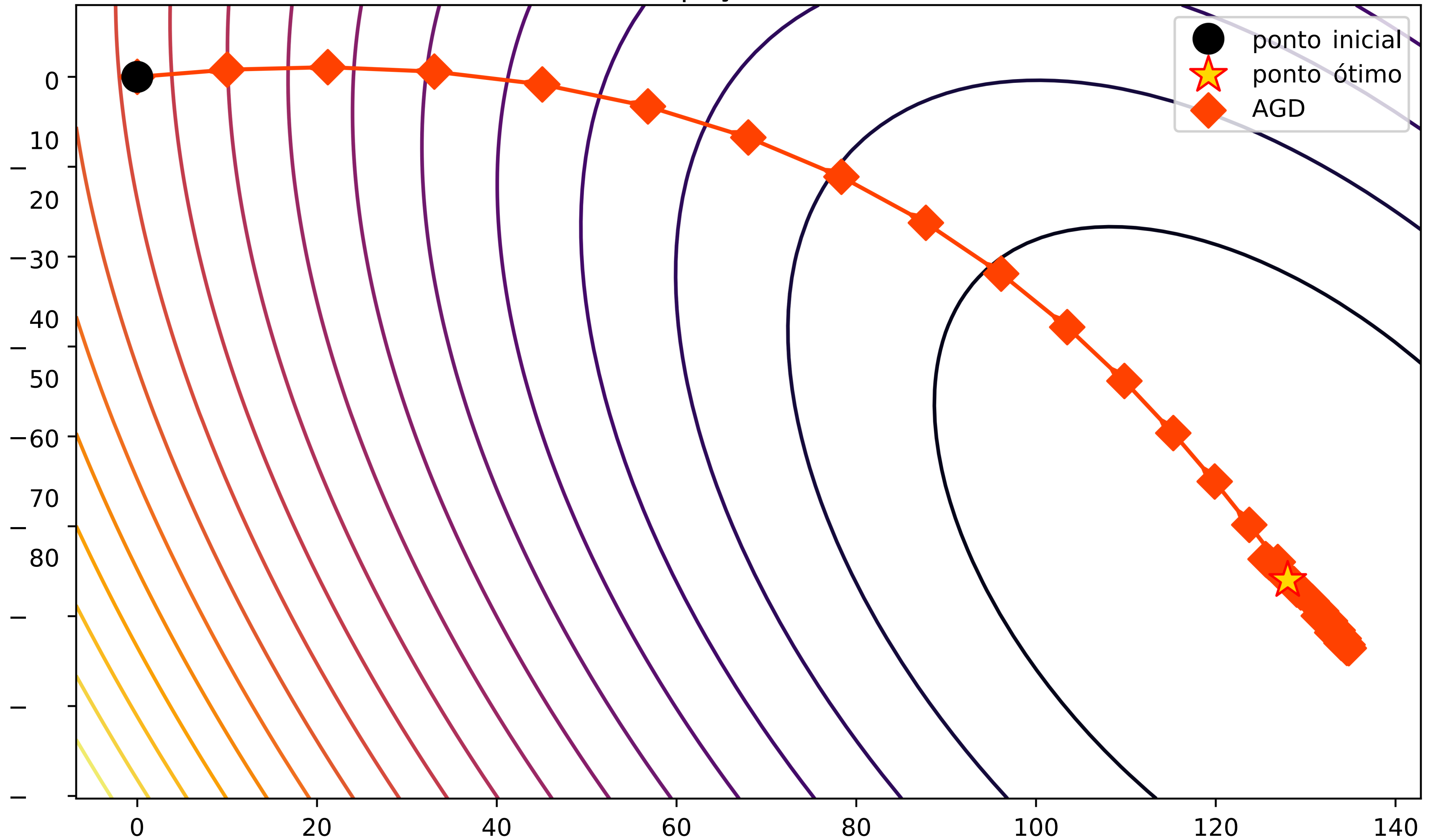


Figura 4.1: Uma iteração do método de Nesterov

Espaço Primal



Accelerated Gradient Descent

$$\begin{cases} x_{k+1} &= y_k - \eta \nabla f(y_k) \\ y_k &= x_k + \mu_k(x_k - x_{k-1}) \end{cases}$$

$$\frac{d^2}{dt^2} X(t) + \frac{3}{t} \frac{d}{dt} X(t) + \nabla f(X(t)) = 0$$

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Keywords: Nesterov's accelerated scheme, convex optimization, first-order methods, differential equation, restarting

1. Introduction

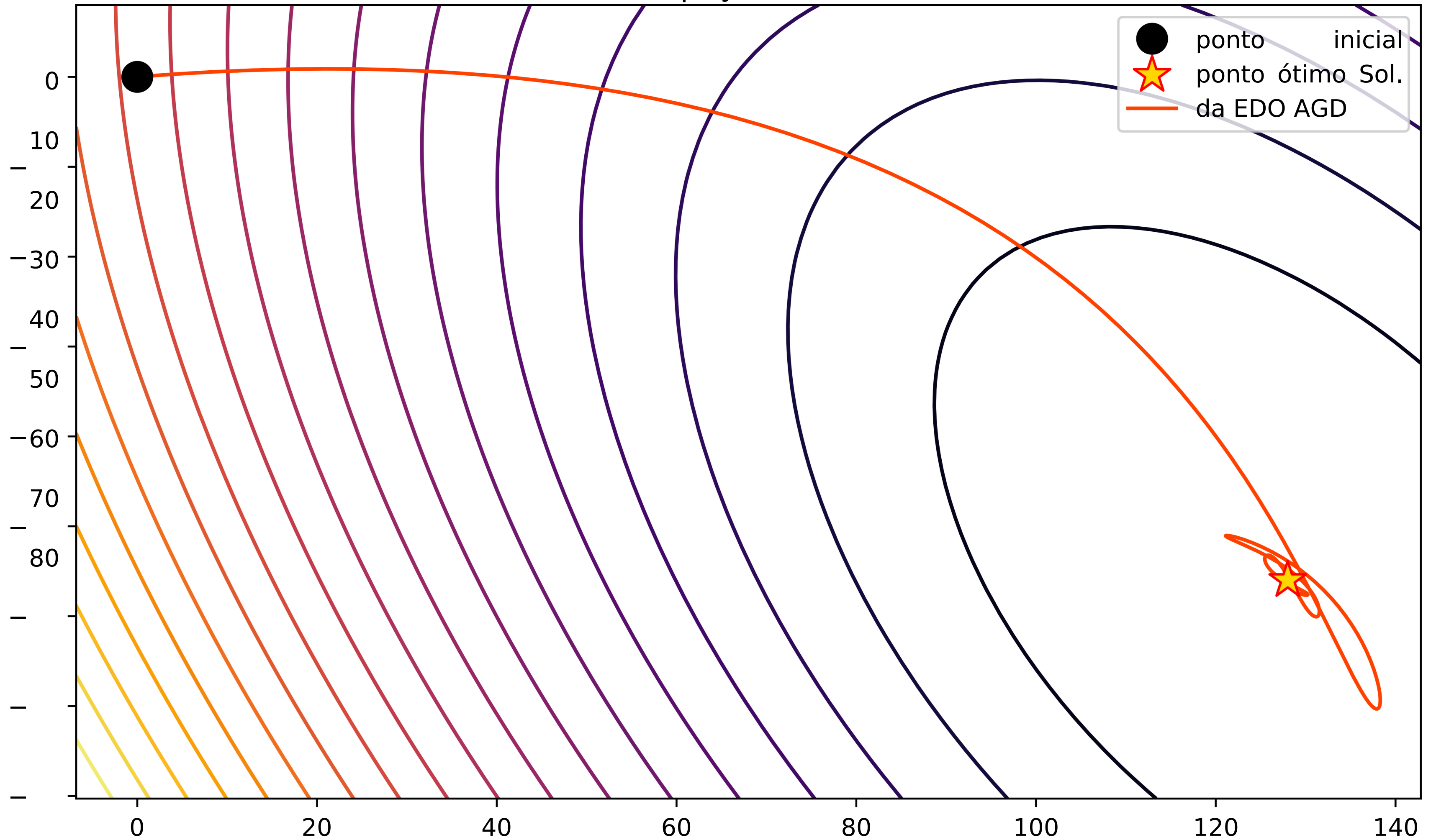
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$$\text{minimize } f(x),$$

where f is a convex function, smooth or non-smooth, and $x \in \mathbb{R}^n$ is the variable. Since Newton, numerous algorithms and methods have been proposed to solve the minimization problem, notably gradient and subgradient descent, Newton's methods, trust region methods, conjugate gradient methods, and interior point methods (see e.g. Polyak, 1987; Boyd and Vandenberghe, 2004; Nocedal and Wright, 2006; Ruszczyński, 2006; Boyd et al., 2011; Shor, 2012; Beck, 2014, for expositions).

First-order methods have regained popularity as data sets and problems are ever increasing in size and, consequently, there has been much research on the theory and practice

Espaço Primal



Accelerated Gradient Descent

Derivação informal da EDO

Ideia:

Pontos críticos:

Soluções:

Accelerated Gradient Descent

Teorema 4.2 (Existência e Unicidade). *Sejam $f \in \mathcal{F}_L$ uma função diferenciável L -Lipschitz e $x_0 \in \mathbb{R}^n$ um ponto qualquer. Então a EDO 4.1 com condições iniciais $X(0) = x_0$ e $\frac{d}{dt}X(0) = \vec{0}$ tem uma única solução $X \in C^2((0, \infty); \mathbb{R}^n) \cap C^1([0, \infty); \mathbb{R}^n)$.*

Teorema 4.3 (Convergência do método de Nesterov para a solução da EDO associada). *Sejam $f \in \mathcal{F}_L$ uma função diferenciável L -Lipschitz e $x_0 \in \mathbb{R}^n$ um ponto qualquer. Sejam ainda $X : \mathbb{R}_+^n \rightarrow \mathbb{R}^n$ a solução da EDO (6.1) com condições iniciais $X(0) = x_0$ e $\frac{d}{dt}\Theta(X(0)) = \vec{0}$ e $\{x_k\}_{\mathbb{N}}$ a sequência de pontos gerada pelo método de Nesterov com ponto inicial x_0 e passo s . Então para qualquer $T \in \mathbb{R}_+$, vale*

$$\lim_{s \rightarrow 0} \sup_{k \in [T/\sqrt{s}]} \|x_k - X(k\sqrt{s})\| = 0 .$$

Mirror Descent

Definição (Divergência de Bregman). *Dada uma função $\Phi : \mathcal{D} \rightarrow \mathbb{R}$ diferenciável e estritamente convexa, a **divergência de Bregman** de x para y com respeito a Φ é definida por*

$$D_{\Phi}(x, y) = \Phi(x) - [\Phi(y) + \nabla\Phi(y)(x - y)].$$

Mirror Descent

$$x_{k+1} = \arg \min_x \{ \eta \nabla f(x_k)(x) + D_{\Phi}(x, x_k) \}$$

$$\begin{cases} x_{k+1} &= (\nabla \Phi)^{-1}(\theta_{k+1}) \\ \theta_{k+1} &= \nabla \Phi(x_k) - \eta \nabla f(x_k) \end{cases}$$

Mirror Descent

Definição (Mapeamento espelho). *Uma função $\Phi : \mathcal{D} \rightarrow \mathbb{R}$ é chamada de **função potencial** se satisfaz as seguintes propriedades:*

- Φ é diferenciável e estritamente convexa em $\mathcal{D}$;
- $\nabla\Phi : \mathcal{D} \rightarrow \mathbb{R}^{n*}$ é sobrejetiva, ou seja, $\nabla\Phi(\mathcal{D}) = \mathbb{R}^{n*}$;
- A norma do gradiente diverge quando x se aproxima da fronteira de $\mathcal{D}$, isto é,

$$\lim_{x \rightarrow \partial\mathcal{D}} \|\nabla\Phi(x)\|_* = \infty.$$

Neste caso, a aplicação $\nabla\Phi$ é denominada **mapeamento espelho** (ou **mirror map**).

Mirror Descent

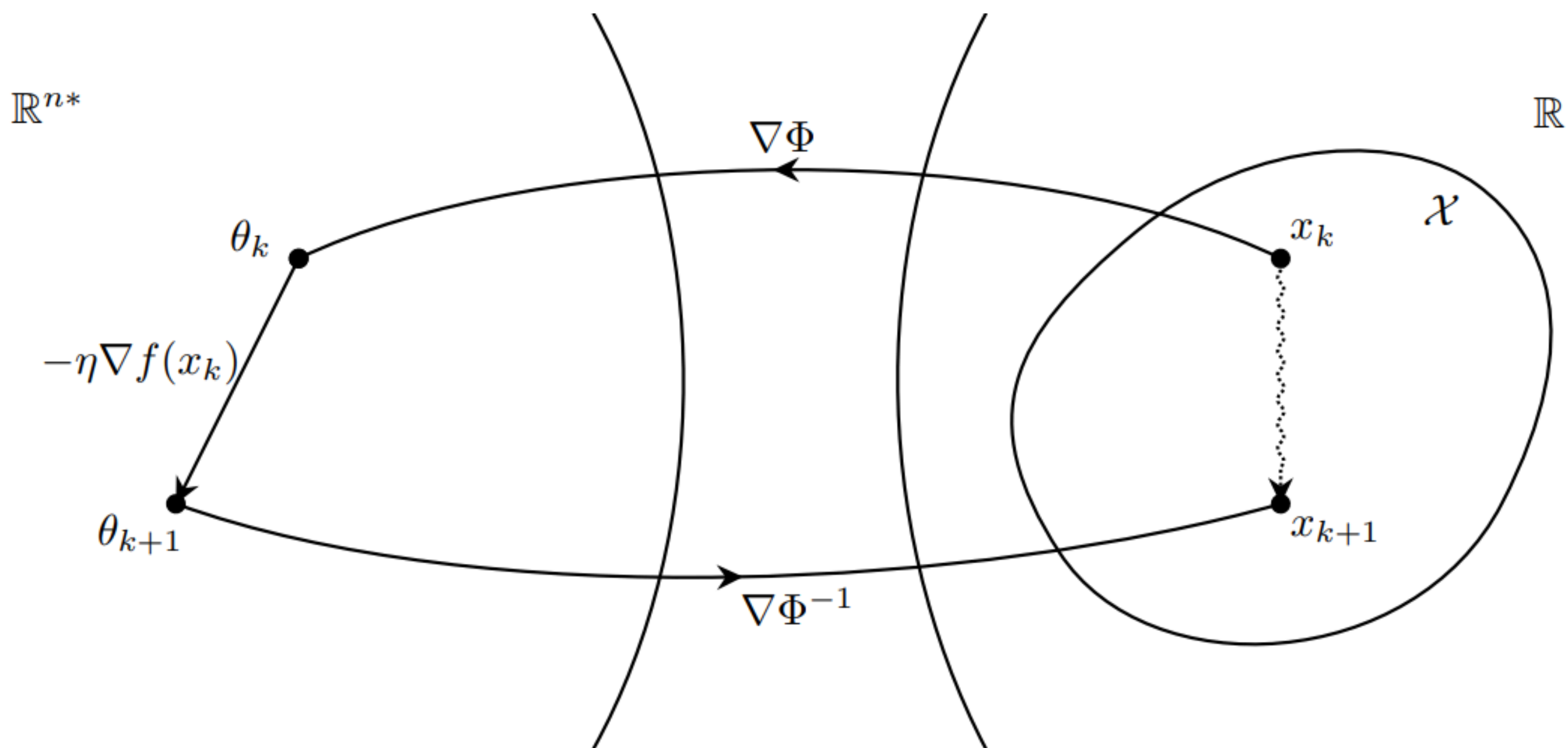
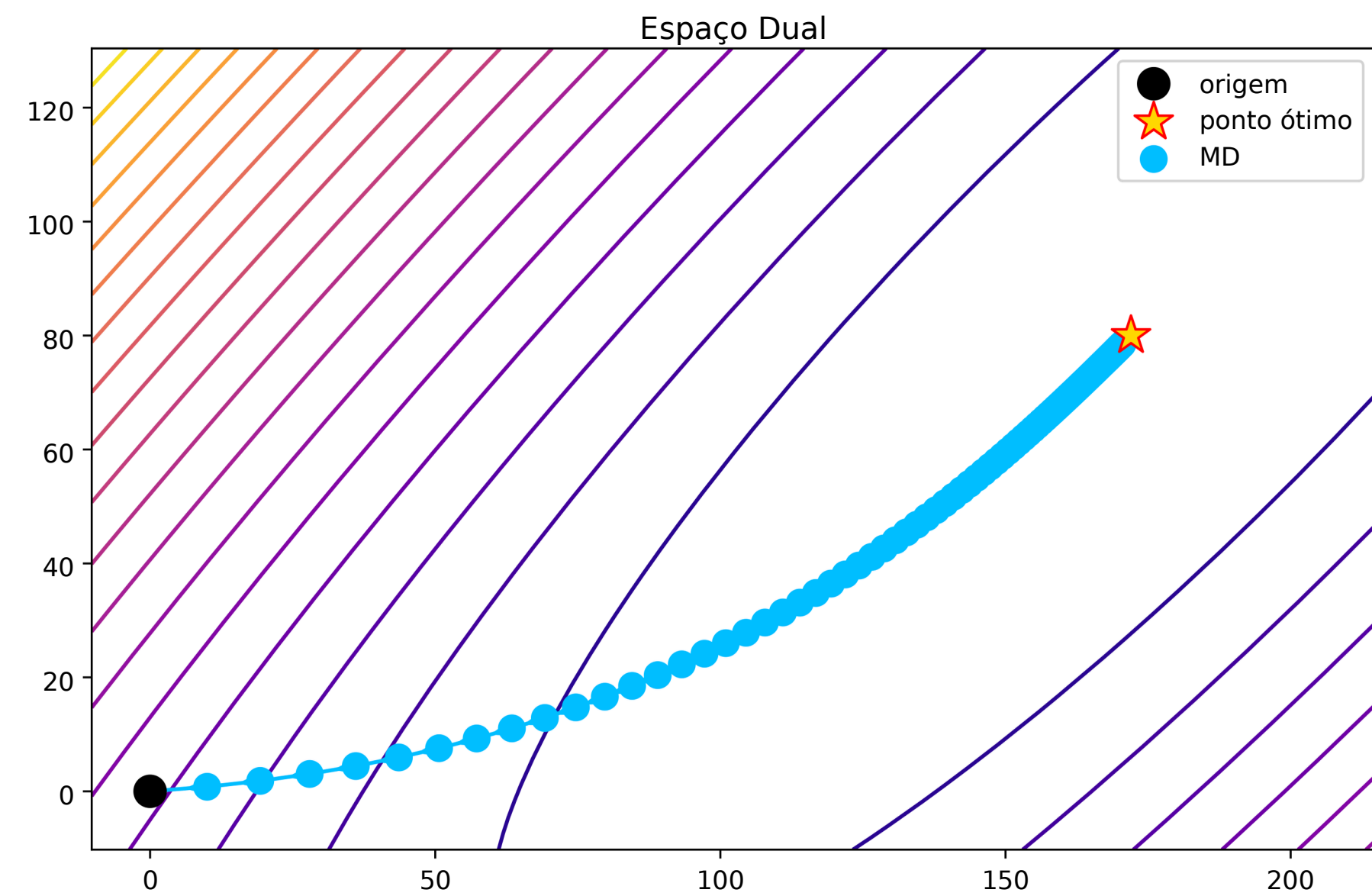
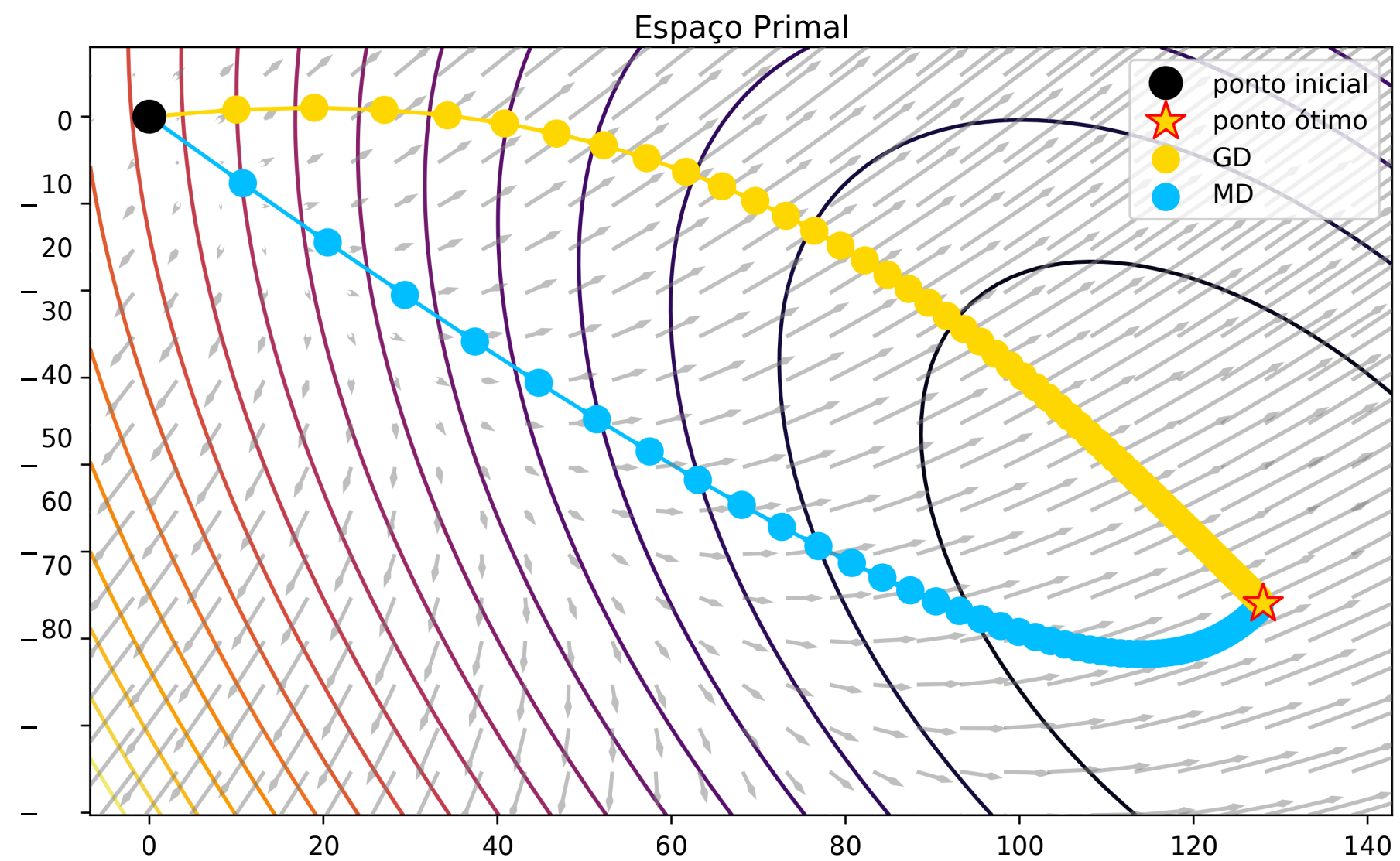


Figura 5.3: Uma iteração do *Mirror Descent*

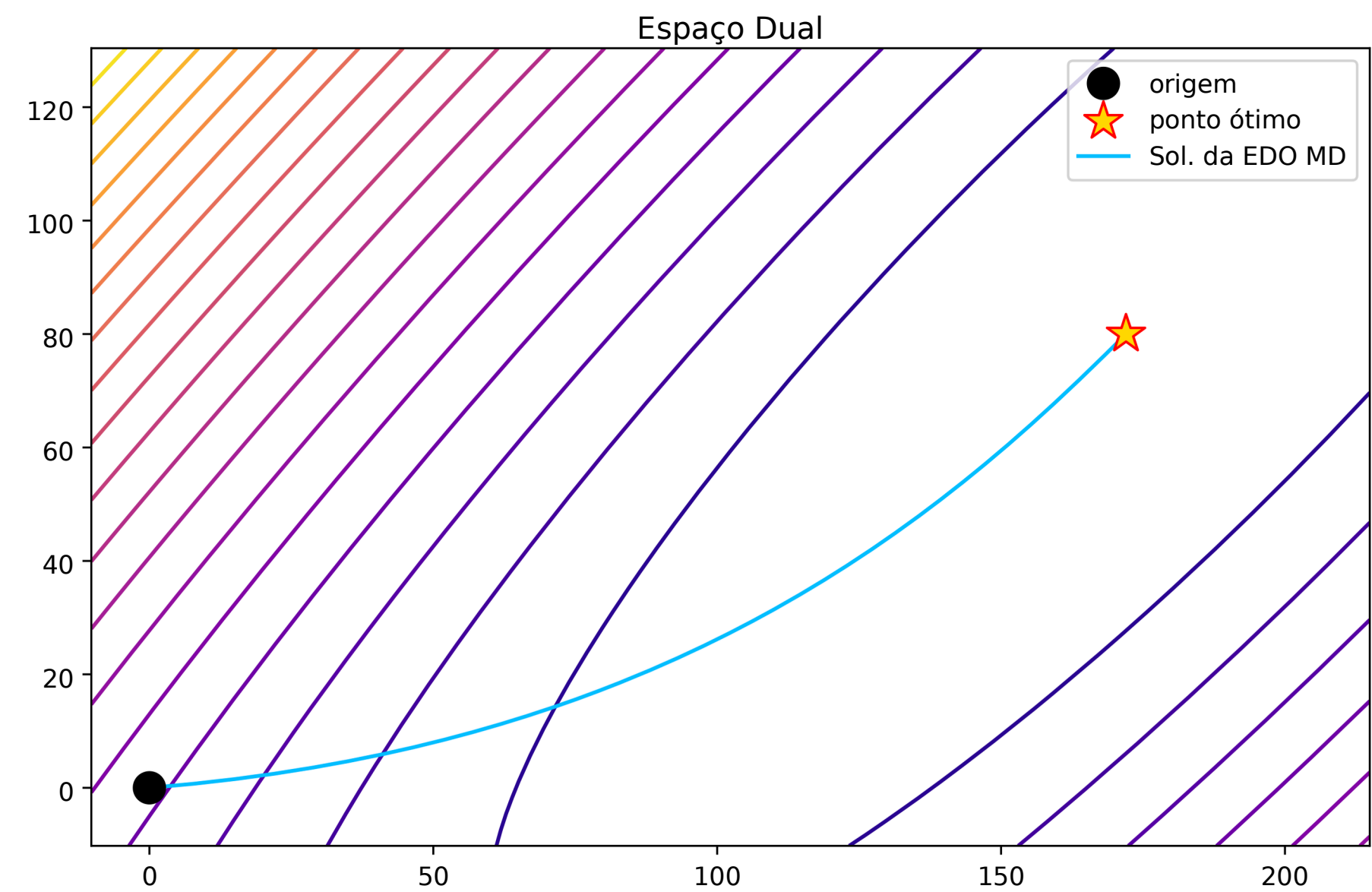
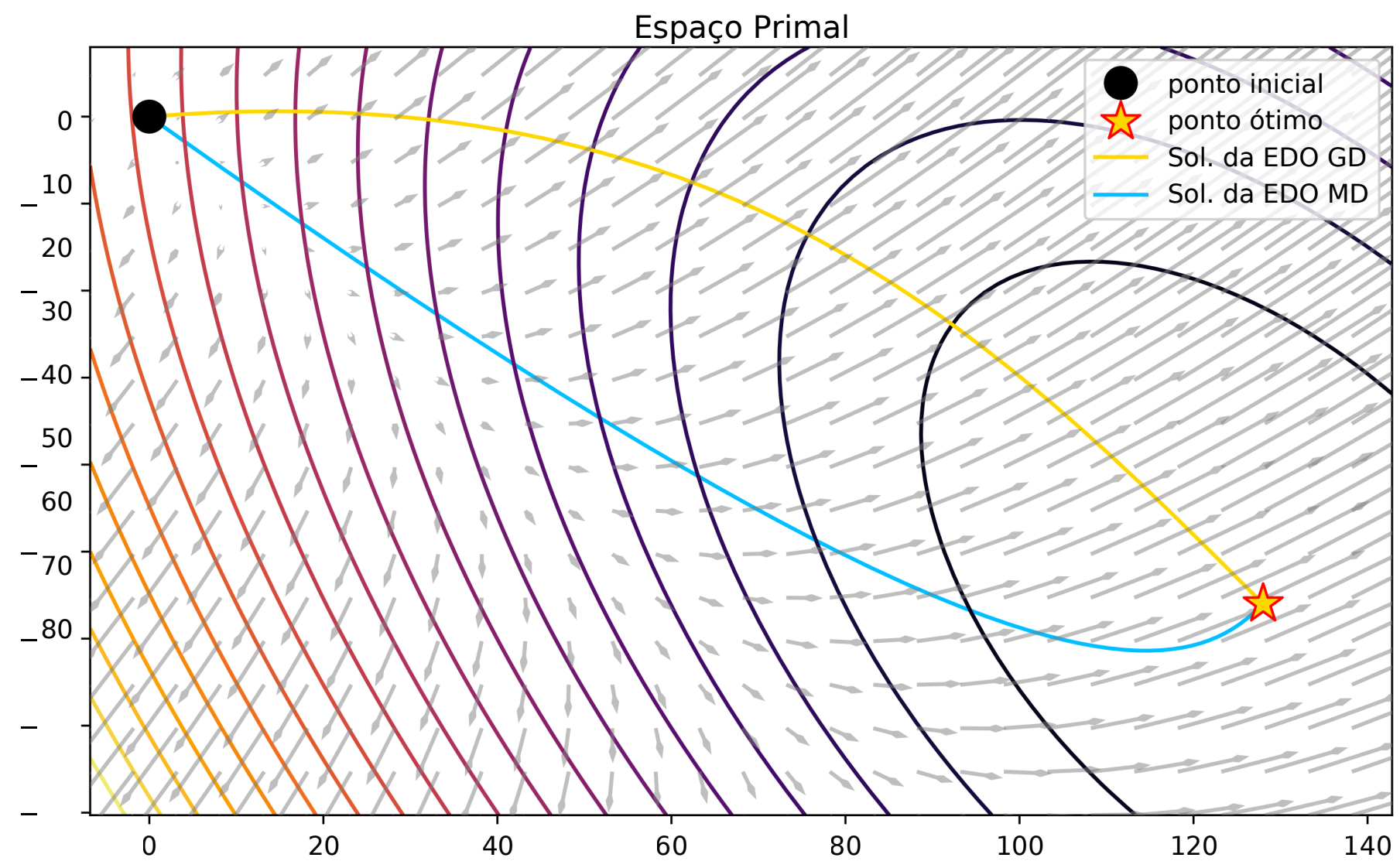


Mirror Descent

$$x_{k+1} = \arg \min_x \{ \eta \nabla f(x_k)(x) + D_{\Phi}(x, x_k) \}$$

$$\begin{cases} x_{k+1} &= (\nabla \Phi)^{-1}(\theta_{k+1}) \\ \theta_{k+1} &= \nabla \Phi(x_k) - \eta \nabla f(x_k) \end{cases}$$

$$\frac{d}{dt} \Theta(X(t)) = -\nabla f(X(t))$$



Mirror Descent

Accelerated Mirror Descent

$$x_{k+1} = x_k - \eta \nabla f(x_k) \quad \curvearrowright \quad \begin{cases} x_{k+1} &= y_k - \eta \nabla f(y_k) \\ y_k &= x_k + \mu_k (x_k - x_{k-1}) \end{cases}$$

$$\begin{cases} x_{k+1} &= (\nabla \Phi)^{-1}(\theta_{k+1}) \\ \theta_{k+1} &= \nabla \Phi(x_k) - \eta \nabla f(x_k) \end{cases} \quad ??$$

Accelerated Mirror Descent

$$\begin{cases} x_{k+1} &= (\nabla \Phi)^{-1}(\theta_{k+1}) \\ \theta_{k+1} &= \nabla \Phi(x_k) - \eta \nabla f(x_k) \end{cases}$$

$$\begin{cases} x_{k+1} &= (\nabla \Phi)^{-1}(\theta_{k+1}) \\ \theta_{k+1} &= \xi_k - \eta \nabla f(y_k) \\ \xi_k &= \nabla \Phi(y_k) \\ y_k &= x_k + \mu_k(x_k - x_{k-1}) \end{cases}$$



$$\begin{cases} x_{k+1} &= (\nabla \Phi)^{-1}(\theta_{k+1}) \\ \theta_{k+1} &= \xi_k - \eta \nabla f(y_k) \\ y_k &= (\nabla \Phi)^{-1}(\xi_k) \\ \xi_k &= \theta_k + \mu_k(\theta_k - \theta_{k-1}) \end{cases}$$

Accelerated Mirror Descent

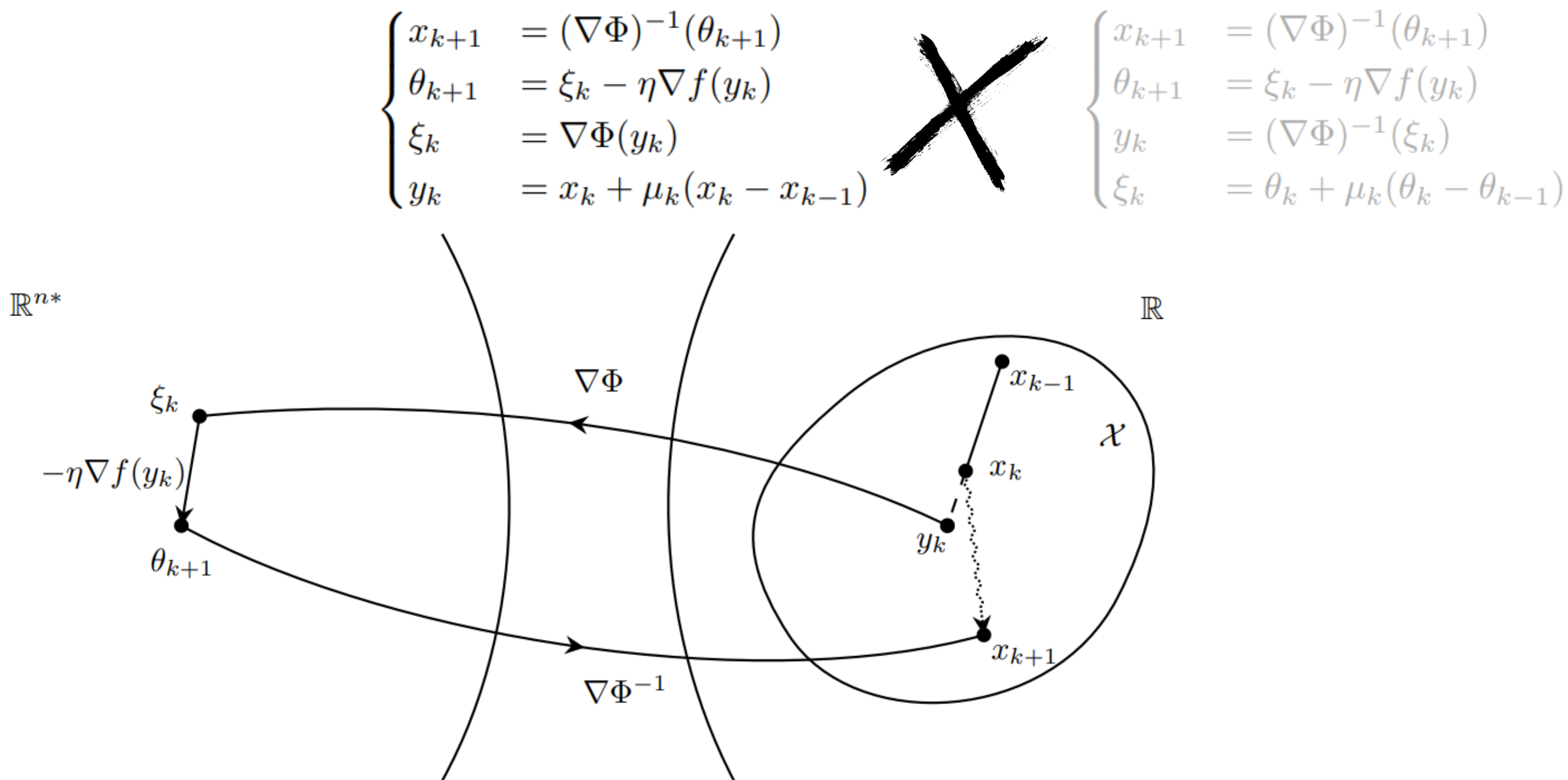


Figura 6.1: Uma iteração do método do P-AMD

Accelerated Mirror Descent

$$\left\{ \begin{array}{l} x_{k+1} = (\nabla\Phi)^{-1}(\theta_{k+1}) \\ \theta_{k+1} = \xi_k - \eta\nabla f(y_k) \\ \xi_k = \nabla\Phi(y_k) \\ y_k = x_k + \mu_k(x_k - x_{k-1}) \end{array} \right. \quad \times \quad \left\{ \begin{array}{l} x_{k+1} = (\nabla\Phi)^{-1}(\theta_{k+1}) \\ \theta_{k+1} = \xi_k - \eta\nabla f(y_k) \\ y_k = (\nabla\Phi)^{-1}(\xi_k) \\ \xi_k = \theta_k + \mu_k(\theta_k - \theta_{k-1}) \end{array} \right.$$

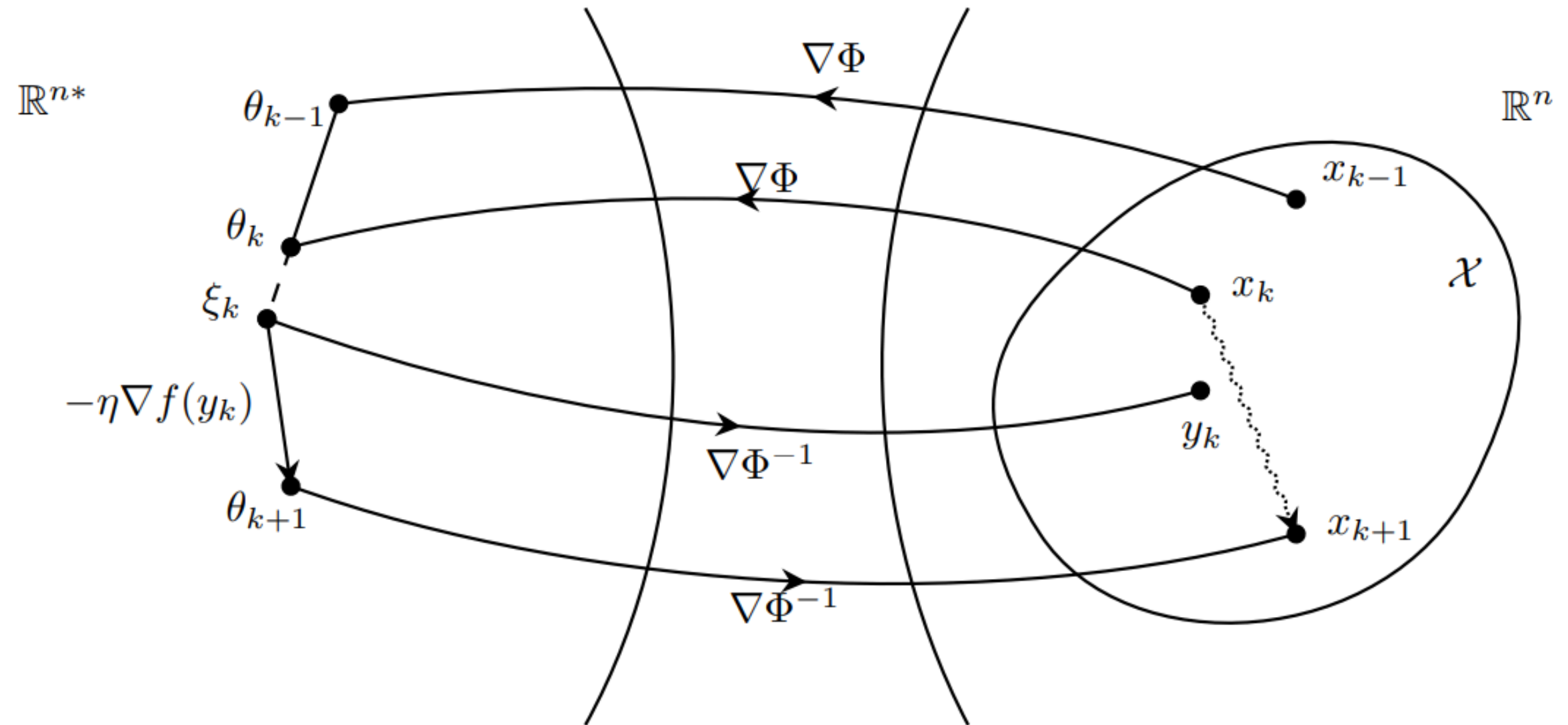
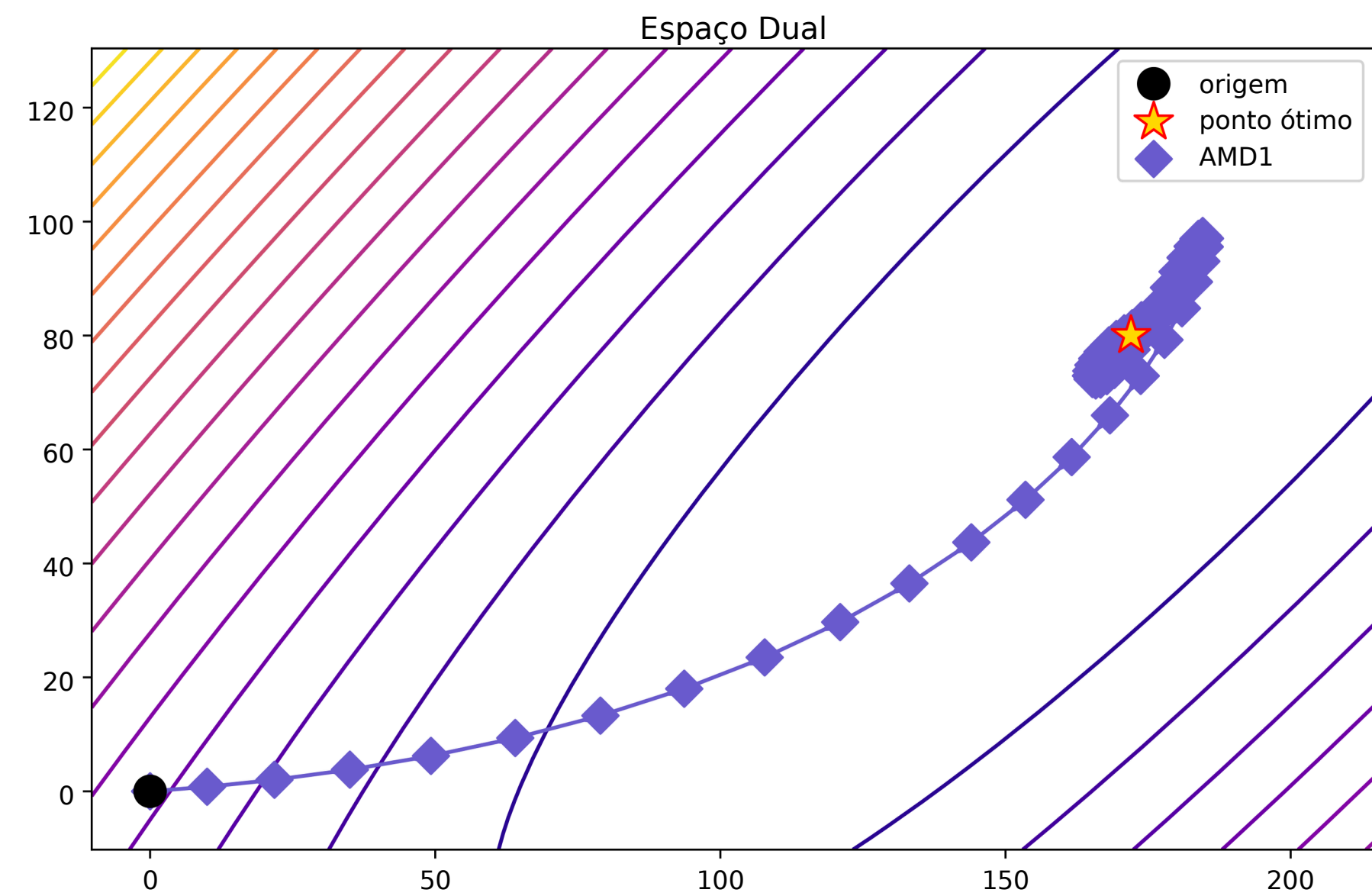
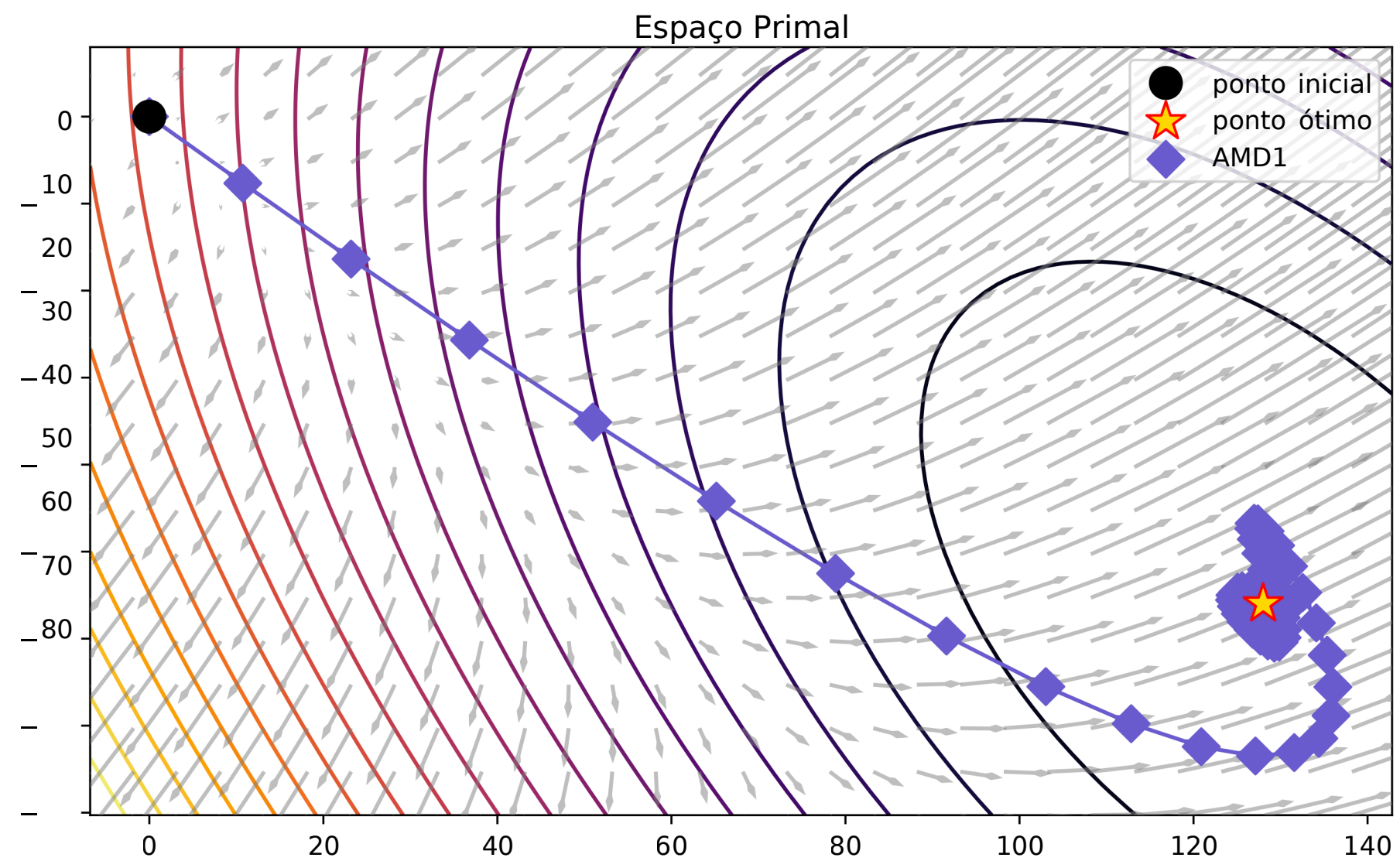


Figura 6.2: Uma iteração do método do D-AMD



Accelerated Mirror Descent

$$\begin{cases} x_{k+1} &= y_k - \eta \nabla f(y_k) \\ y_k &= x_k + \mu_k(x_k - x_{k-1}) \end{cases}$$



$$\frac{d^2}{dt^2}X(t) + \frac{3}{t} \frac{d}{dt}X(t) + \nabla f(X(t)) = 0$$

$$\begin{cases} x_{k+1} &= (\nabla \Phi)^{-1}(\theta_{k+1}) \\ \theta_{k+1} &= \xi_k - \eta \nabla f(y_k) \\ y_k &= (\nabla \Phi)^{-1}(\xi_k) \\ \xi_k &= \theta_k + \mu_k(\theta_k - \theta_{k-1}) \end{cases}$$

Accelerated Mirror Descent

$$\begin{cases} x_{k+1} &= y_k - \eta \nabla f(y_k) \\ y_k &= x_k + \mu_k(x_k - x_{k-1}) \end{cases}$$



$$\frac{d^2}{dt^2} X(t) + \frac{3}{t} \frac{d}{dt} X(t) + \nabla f(X(t)) = 0$$

$$\begin{cases} x_{k+1} &= (\nabla \Phi)^{-1}(\theta_{k+1}) \\ \theta_{k+1} &= \xi_k - \eta \nabla f(y_k) \\ y_k &= (\nabla \Phi)^{-1}(\xi_k) \\ \xi_k &= \theta_k + \mu_k(\theta_k - \theta_{k-1}) \end{cases}$$



$$\frac{d^2}{dt^2} \Theta(X(t)) + \frac{3}{t} \frac{d}{dt} \Theta(X(t)) + \nabla f(X(t)) = 0$$

Accelerated Mirror Descent

$$\frac{d^2}{dt^2}\Theta(X(t)) + \frac{3}{t} \frac{d}{dt}\Theta(X(t)) + \nabla f(X(t)) = 0$$

Teorema 7.1. *Sejam $f \in \mathcal{F}_L$ uma função diferenciável com gradiente ∇f L -Lipschitz, Φ uma função ν -fortemente convexa e diferenciável e $x_0 \in \mathbb{R}^n$ um ponto qualquer. Então a EDO (6.1) com condições iniciais $X(0) = x_0$ e $\frac{d}{dt}\Theta(X(0)) = \vec{0}$ tem uma única solução $X \in C^2((0, \infty); \mathbb{R}^n) \cap C^1([0, \infty); \mathbb{R}^n)$.*

Teorema 7.16. *Sejam $f \in \mathcal{F}_L$ uma função diferenciável L -Lipschitz com um mínimo global $y^* \in \mathbb{R}^n$ e $x_0 \in \mathbb{R}^n$ um ponto qualquer. Sejam ainda $X : \mathbb{R}_+^n \rightarrow \mathbb{R}^n$ a solução da EDO (6.1) com condições iniciais $X(0) = x_0$ e $\frac{d}{dt}\Theta(X(0)) = \vec{0}$ e $\{x_k\}_{\mathbb{N}}$ a sequência de pontos gerada pelo AMD com ponto inicial x_0 e passo s . Então para qualquer $T \in \mathbb{R}_+$,*

$$\lim_{s \rightarrow 0} \sup_{k \in [T/\sqrt{s}]} \|x_k - X(k\sqrt{s})\| = 0 .$$

Existência

Ideia: criar uma sequência de EDOs convergindo para a EDO original nas quais podemos aplicar o teorema de Picard e depois tomar uma solução que corresponde a um limite de uma subsequência de soluções dessas EDOs.

Pontos críticos:

Soluções:

Existência

$$\frac{d^2}{dt^2}\Theta(X(t)) + \frac{3}{\max(\delta, t)} \frac{d}{dt}\Theta(X(t)) + \nabla f(X(t)) = 0$$

Unicidade

Convergência